# Web Programming Using Python and JavaScript Lab
# [SEC035]
# Lab Report Submitted to

## Bachelor of Technology
## In
## Computer Science Engineering
## SEMESTER-IV


## Submitted by
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## SECTION-B

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## K. R. MANGALAM UNIVERSITY

# EXPERIMENT–1

**AIM-** To design and develop a personal portfolio website using HTML, CSS, and JavaScript that displays personal information, skills, projects, services, and contact details in a structured, attractive, and interactive manner.

## Technologies Used-

- HTML – For creating the structure of the website
- CSS – For styling, layout, colors, and responsiveness
- JavaScript – For adding interactivity like buttons, show/hide, and navigation effects
- Web Browser – To run and test the website

## Explanation-

**HTML Structure**

- index.html contains the main structure of the portfolio.
- Semantic tags such as <header>, <nav>, <section>, and <footer> are used.
- Separate HTML pages are created for different services like frontend,
- database, and Java.

**CSS Styling**

- style.css is linked externally to all HTML files.
- CSS is used for:
    - Layout alignment
    - Background colors and fonts
    - Spacing using margin and padding
    - Styling buttons, cards, and sections

**JavaScript Functionality**

- script.js is used to:
    - Handle button clicks
    - Add interactive effects
    - Improve navigation experience

## HTML CODE:

resume.html > html > body > section#contact.contact.reveal

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Shikha | Portfolio</title>
    <link rel="stylesheet" href="re.css">
</head>
<body>

<!-- NAVBAR -->
<header class="navbar">
    <div class="logo">SJ</div>
    <nav>
        <a href="#about">About</a>
        <a href="#skills">Skills</a>
        <a href="#projects">Projects</a>
        <a href="#services">Services</a>
        <a href="#contact">Contact</a>
    </nav>
</header>

<!-- HERO / ABOUT -->
<section id="about" class="hero reveal">
    <div class="hero-img">
    <img src="shikha.jpeg" alt="Shikha">
</div>


    <div class="hero-text">
        <h1>Hey, I'm <span>Shikha</span></h1>
        <p>
            B.Tech CSE student exploring core CS concepts and web development.
            Learning by building, breaking, and improving every day.
        </p>
    </div>
</section>
```

```html
<!-- SKILLS -->
<section id="skills" class="skills reveal">
    <h2>Skills</h2>
    <div class="skill-box">
        <h1 style="font-size: 24px;">Technologies and Frameworks</h1>
        <span>Git</span>
        <span>GitHub</span>
        <span>VS Code</span>
        <span>Netlify</span>
        <h1 style="font-size: 24px; margin-top: 20px;">Programming Languages</h1>
        <span>HTML</span>
        <span>CSS</span>
        <span>JavaScript</span>
        <span>Python</span>
        <span>C++</span>
        <h1 style="font-size: 24px; margin-top: 20px;">Core Concepts</h1>
        <span>Data Structures</span>
        <span>Algorithms</span>
        <span>Database Management Systems</span>
        <h1 style="font-size: 24px; margin-top: 20px;">Soft Skills</h1>
        <span>Team Collaboration</span>
        <span>Time Management</span>
        <span>Active Listening</span>
        <span>Problem Solving</span>
    </div>
</section>
```

```html
<!-- PROJECTS -->
<section id="projects" class="projects reveal">
    <h2>Projects</h2>

    <div class="project-grid">
        <div class="project-card" onclick="openProject('project1.html')">
            <h3>InStock - Inventory Management System</h3>
            <p>A web application for managing inventory in retail stores.</p>
        </div>

        <div class="project-card" onclick="openProject('project2.html')">
            <h3>Recipe Haven</h3>
            <p>A web application for managing and sharing recipes.</p>
        </div>

    </div>
</section>
```

```html
<body>

<section id="contact" class="contact reveal">
    <h2>Contact Me</h2>

    <div class="contact-links">
        <a href="mailto:jha89719@gmail.com">📧 Email</a>
        <a href="https://linkedin.com" target="_blank">💼 LinkedIn</a>
    </div>

    <p class="msg-trigger" onclick="openModal()">Send me a message 💬</p>

    <div class="modal" id="msgModal">
        <div class="modal-box">
            <span class="close" onclick="closeModal()">×</span>

            <h3>How would you like to send your message?</h3>

            <div class="modal-options">
                <button type="button" onclick="showAnonymous()">Send Anonymously</button>
                <button type="button" onclick="showNormal()">Send With Name</button>
            </div>

            <form class="form anon-form">
                <textarea placeholder="Your Message" required></textarea>
                <button type="submit">Send</button>
            </form>

            <form class="form normal-form">
                <input type="text" placeholder="Your Name" required>
                <input type="email" placeholder="Your Email" required>
                <textarea placeholder="Your Message" required></textarea>
                <button type="submit">Send</button>
            </form>
        </div>
    </div>
```

```html
                </form>
            </div>
        </div>
    </section>



    <footer>
        <p>© 2026 Shikha.</p>
    </footer>

    <script src="on.js"></script>

    </body>
    </html>
```

## CSS CODE:

```css
* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
}

html {
    scroll-behavior: smooth;
}

body {
    font-family: 'Segoe UI', sans-serif;
    background: #f8fafc;
    color: #111827;
}

/* NAVBAR */
.navbar {
    position: fixed;
    top: 0;
    width: 100%;
    padding: 15px 60px;
    display: flex;
    justify-content: space-between;
    background: white;
    box-shadow: 0 5px 20px rgba(0,0,0,0.05);
    z-index: 100;
}

.navbar a {
    margin-left: 25px;
    text-decoration: none;
    color: #111827;
}

/* HERO */
.hero {
```

```css
/* HERO */
.hero {
    min-height: 100vh;
    padding: 120px 80px;
    display: flex;
    align-items: center;
    gap: 80px;
}

.hero-img {
    flex: 1;
    display: flex;
    justify-content: center;
}

.hero-img img {
    width: 360px;
    height: 460px;
    object-fit: cover;
    border-radius: 24px;

    transform: rotate(-4deg);
    box-shadow:
        0 25px 50px rgba(0, 0, 0, 0.18),
        0 10px 20px rgba(0, 0, 0, 0.12);

    transition: transform 0.4s ease;
}
.hero-img img:hover {
    transform: rotate(0deg) scale(1.05);
}
.hero-img::before {
    content: "";
    position: absolute;
    width: 360px;
    height: 460px;
```

```css
}
.hero-img {
    position: relative;
}
.hero-text {
    flex: 1;
}


.hero-text h1 {
    font-size: 48px;
}

.hero-text span {
    color: #6366f1;
}

.hero-text p {
    margin-top: 15px;
    font-size: 18px;
    max-width: 500px;
}

/* SECTIONS */
section {
    padding: 100px 80px;
}

h2 {
    font-size: 36px;
    margin-bottom: 40px;
}

/* SKILLS */
.skill-box span {
    display: inline-block;
    background: white;
```

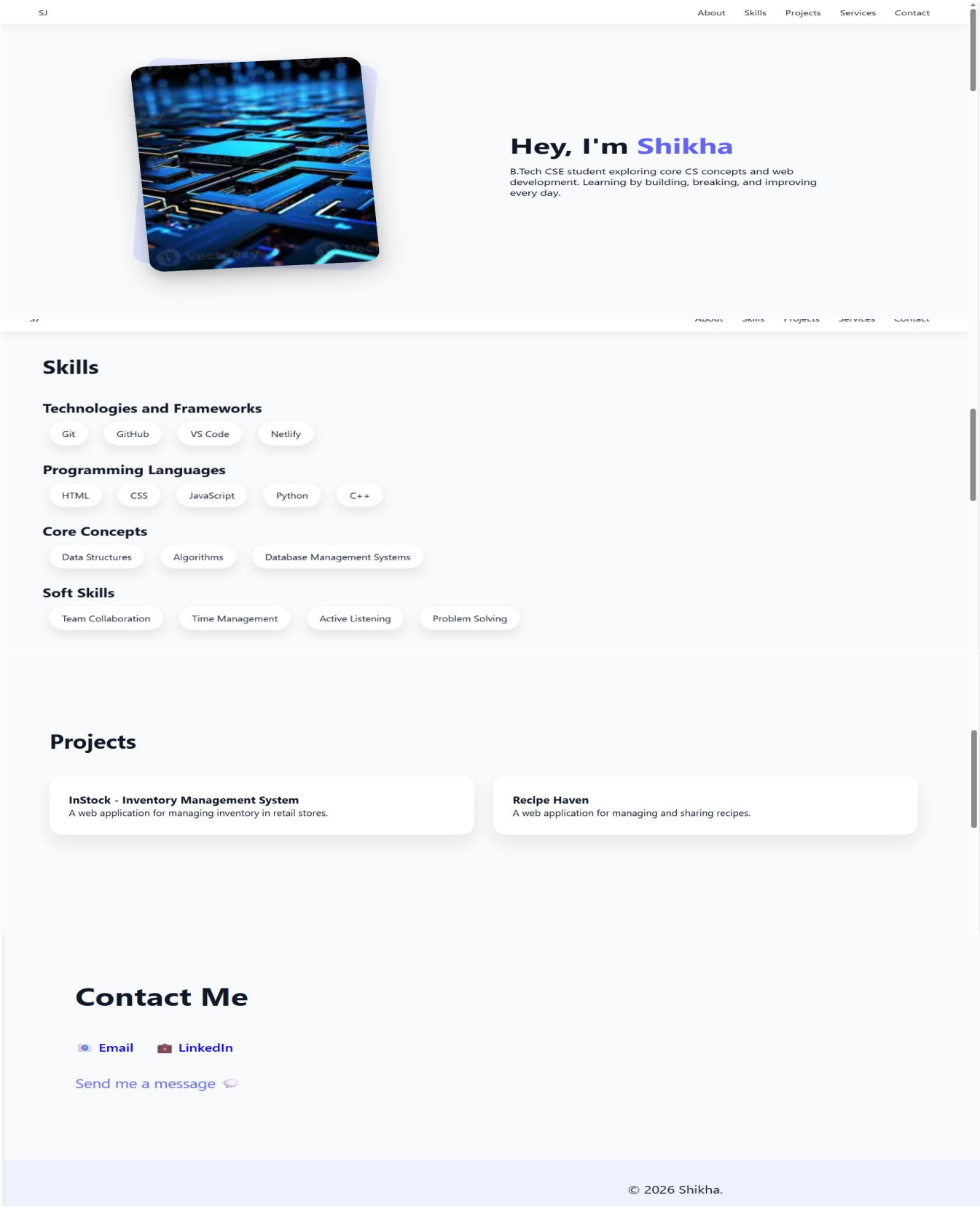
```css
.project-card {
    cursor: pointer;
    transition: 0.3s;
    box-shadow: 0 15px 30px rgba(0,0,0,0.08);
}

.project-card:hover {
    transform: translateY(-8px);
}

/* CONTACT */
.contact-links a {
    margin-right: 20px;
    text-decoration: none;
    font-weight: 500;
}

.msg-line {
    margin: 30px 0 10px;
}

.contact-form input,
.contact-form textarea {
    width: 100%;
    padding: 12px;
    margin-bottom: 15px;
}

.contact-form button {
    padding: 12px 25px;
    background: #6366f1;
    color: white;
    border: none;
    border-radius: 8px;
    cursor: pointer;
```

```
.anon {
    gap: 8px;
    margin-bottom: 15px;
}
.msg-trigger {
    margin-top: 30px;
    font-size: 18px;
    color: #6366f1;
    cursor: pointer;
}

/* MODAL BACKGROUND */
.modal {
    position: fixed;
    inset: 0;
    background: rgba(0,0,0,0.4);
    display: none;
    align-items: center;
    justify-content: center;
    z-index: 999;
}

/* MODAL BOX */
.modal-box {
    background: white;
    padding: 30px;
    width: 400px;
    border-radius: 16px;
    position: relative;
    animation: scaleIn 0.3s ease;
}

@keyframes scaleIn {
    from { transform: scale(0.9); opacity: 0; }
    to { transform: scale(1); opacity: 1; }
}
```

**JAVASCRIPT CODE:**

```javascript
document.addEventListener('DOMContentLoaded', () => {

    // ---------- PROJECT OPEN ----------
    window.openProject = function (page) {
        window.location.href = page;
    };

    // ---------- SCROLL REVEAL ----------
    const reveals = document.querySelectorAll('.reveal');

    window.addEventListener('scroll', () => {
        reveals.forEach(section => {
            const top = section.getBoundingClientRect().top;
            if (top < window.innerHeight - 100) {
                section.classList.add('active');
            }
        });
    });

    // ---------- CLICK HIGHLIGHT ----------
    const sections = document.querySelectorAll('.reveal');

    sections.forEach(section => {
        section.addEventListener('click', () => {

            // remove active-click from all
            sections.forEach(sec => sec.classList.remove('active-click'));

            // add to clicked one
            section.classList.add('active-click');
        });
    });

    // ---------- MODAL ----------
    const modal = document.getElementById("msgModal");
    const anonForm = document.querySelector(".anon-form");
    const normalForm = document.querySelector(".normal-form");

    window.openModal = function () {
        modal.style.display = "flex";
    };

    window.closeModal = function () {
        modal.style.display = "none";
        anonForm.style.display = "none";
        normalForm.style.display = "none";
    };

    window.showAnonymous = function () {
        anonForm.style.display = "block";
        normalForm.style.display = "none";
    };

    window.showNormal = function () {
        normalForm.style.display = "block";
        anonForm.style.display = "none";
    };

});
// ---------- CLICK GLOW FOR CARDS ----------
const cards = document.querySelectorAll('.project-card, .service-card');

cards.forEach(card => {
    card.addEventListener('click', () => {

        cards.forEach(c => c.classList.remove('active-click'));

        card.classList.add('active-click');
    });
});
```

**OUTPUT:**




**RESULT:** A fully functional, professional-looking personal portfolio website was successfully created using HTML, CSS, and JavaScript.

# EXPERIMENT-2

**AIM-** To design and develop a professional company landing page named Bright Solution using HTML and CSS, displaying company information, services, testimonials, and contact details in a well-structured and visually appealing manner.

**Technologies Used-** HTML – To create the structure of the landing page - CSS – To design layout, colors, fonts, and overall appearance - Web Browser – To test and run the website

## Explanation-

**HTML Structure**

• index.html contains the main structure of the landing page.
• Semantic tags like <header>, <section>, and <footer> are used.
• Sections included:

- Header / Hero Section
- About / Introduction
- Services
- Testimonials
- Contact Information

• Separate HTML pages are created for each service (Web Development, Database, IT Consulting).

**CSS Styling**

• style.css is linked externally to all pages.
• CSS is used to:

- Style layout and spacing
- Apply background colors and fonts
- Align sections and create service cards
- Improve readability and visual appeal

## HTML CODE:

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Bright Solution | Home</title>
    <link rel="stylesheet" href="company.css">
</head>
<body>

<header>
    <div class="container">
        <h1>Bright Solution</h1>
        <p>Your Partner in Digital Growth</p>
        <a href="#contact" class="btn">Get in Touch</a>
    </div>
</header>

<nav>
    <ul>
        <li><a href="index.html">Home</a></li>
        <li><a href="#about">About</a></li>
        <li><a href="#services">Services</a></li>
        <li><a href="#testimonials">Testimonials</a></li>
        <li><a href="#contact">Contact</a></li>
    </ul>
</nav>

<section id="about" class="section">
    <div class="container">
        <h2>About Us</h2>
        <p>Bright Solution is a professional IT company providing innovative technology services to businesses worldwide.
        We specialize in modern web development, database solutions, and IT consulting to help organizations grow efficiently.</p>
    </div>
</section>

<section id="services" class="section light">
    <div class="container">
```

```html
<section id="services" class="section light">
    <div class="container">
        <h2>Our Services</h2>
        <div class="card-container">

            <div class="card">
                <h3>Web Development</h3>
                <p>Modern and responsive website development using latest technolog
                <a href="company2.html" class="btn-small">Read More</a>
            </div>

            <div class="card">
                <h3>Database Solutions</h3>
                <p>Secure and scalable database design and management services.</p>
                <a href="company3.html" class="btn-small">Read More</a>
            </div>

            <div class="card">
                <h3>IT Consulting</h3>
                <p>Strategic IT consulting to optimize your business processes.</p>
                <a href="company4.html" class="btn-small">Read More</a>
            </div>

        </div>
    </div>
</section>

<section id="testimonials" class="section">
    <div class="container">
        <h2>What Our Clients Say</h2>

        <div class="testimonial">
            <p>"Bright Solution transformed our online presence completely!"</p>
            <h4>- Rahul Mehta</h4>
        </div>
```

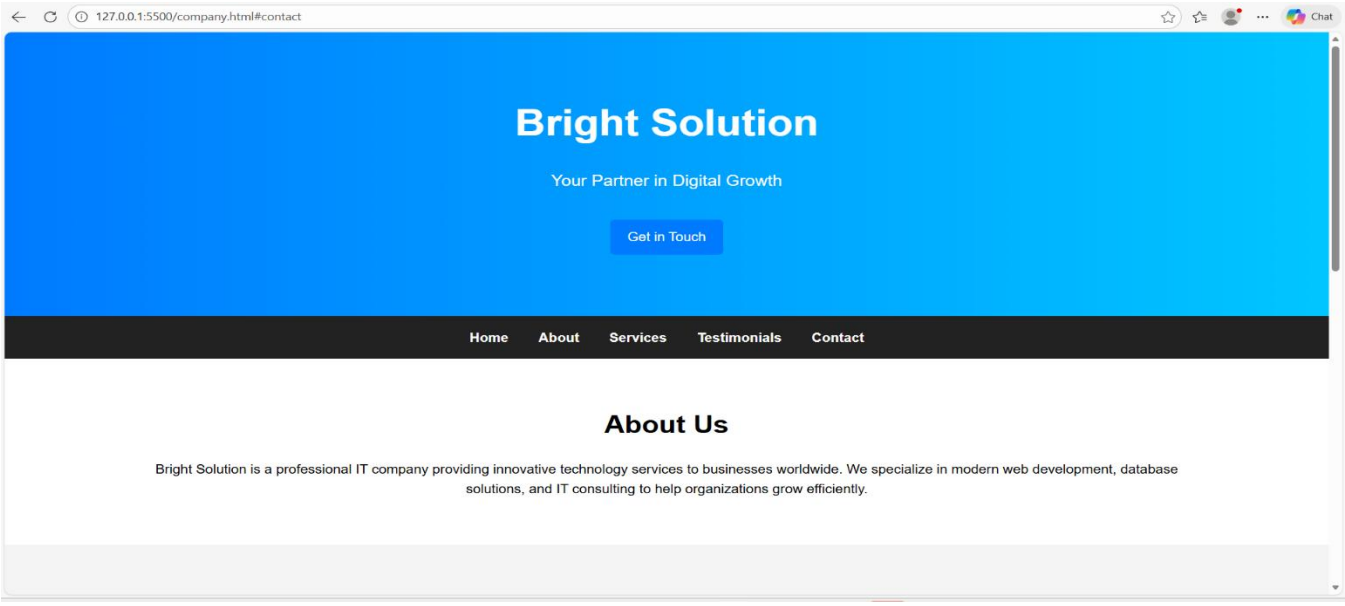
```html
        <div class="testimonial">
            <p>"Professional team and excellent support. Highly recommended."</p>
            <h4>- Sneha Kapoor</h4>
        </div>

    </div>
</section>

<section id="contact" class="section light">
    <div class="container">
        <h2>Contact Us</h2>
        <p>Email: info@brightsolution.com</p>
        <p>Phone: +91 9876543210</p>
        <p>Address: New Delhi, India</p>
    </div>
</section>

<footer>
    <p>© 2026 Bright Solution. All Rights Reserved.</p>
</footer>

</body>
</html>
```

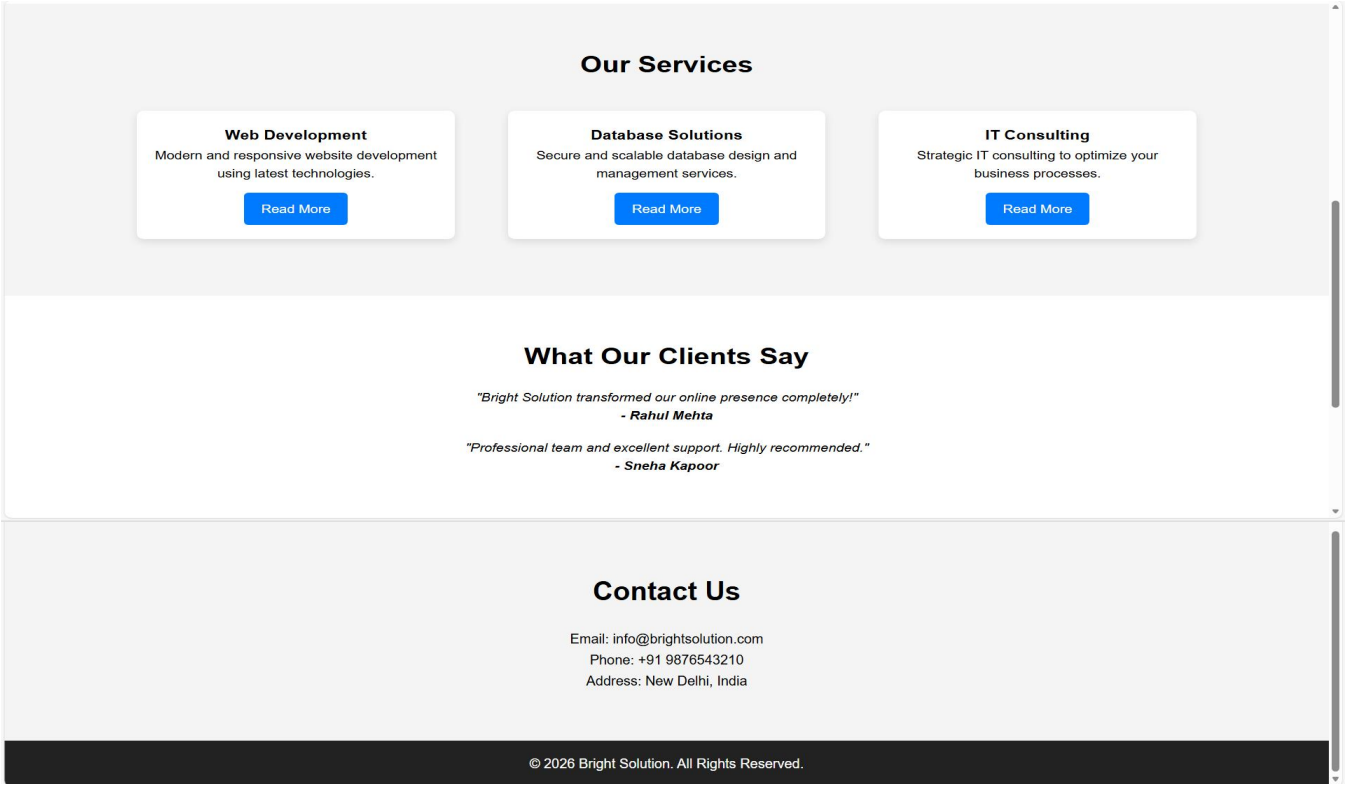
**CSS**
**CODE:**

```
mpany.css > ...
* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
    font-family: Arial, sans-serif;
}

body {
    line-height: 1.6;
}

.container {
    width: 80%;
    margin: auto;
}

header {
    background: linear-gradient(to right, #007bff, #00c6ff);
    color: white;
    text-align: center;
    padding: 80px 20px;
}

header h1 {
    font-size: 48px;
}

header p {
    font-size: 20px;
    margin: 20px 0;
}

nav {
    background: #222;
    padding: 15px 0;
}
nav ul li a {
    font-weight: bold;
}

nav ul li a:hover {
    color: #00c6ff;
}

.section {
    padding: 60px 0;
    text-align: center;
}

.section h2 {
    margin-bottom: 20px;
    font-size: 32px;
}

.light {
    background: #f4f4f4;
}

.card-container {
    display: flex;
    justify-content: space-between;
    margin-top: 40px;
    flex-wrap: wrap;
}

.card {
    background: white;
    padding: 20px;
    width: 30%;
    box-shadow: 0 4px 10px rgba(0,0,0,0.1);
    border-radius: 8px;
    margin-bottom: 20px;
```

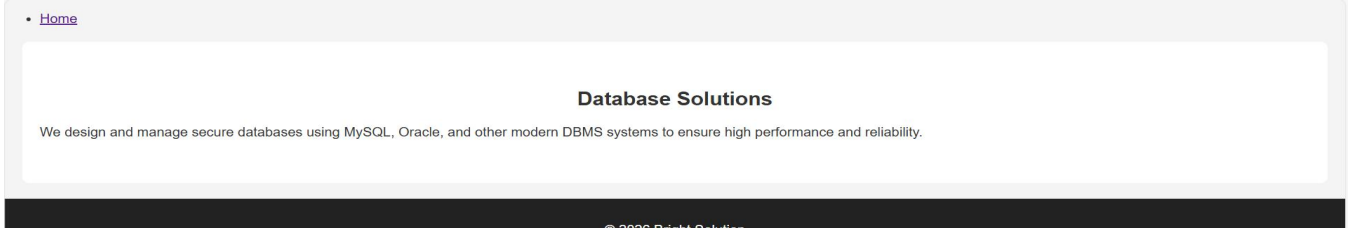
```css
.btn, .btn-small {
    display: inline-block;
    background: #007bff;
    color: white;
    padding: 10px 20px;
    text-decoration: none;
    border-radius: 5px;
    margin-top: 15px;
}

.btn:hover, .btn-small:hover {
    background: #0056b3;
}

.testimonial {
    margin: 20px 0;
    font-style: italic;
}

footer {
    background: #222;
    color: white;
    text-align: center;
    padding: 15px 0;
}

/* Responsive */
@media(max-width: 768px) {
    .card {
        width: 100%;
    }

    header h1 {
        font-size: 32px;
    }
}
```

# OUTPUT:






## SERVICE SECTION




**RESULT:** A complete and visually appealing company landing page for Bright Solution was successfully developed using HTML and CSS.

# EXPERIMENT–3

**AIM**- To design and develop an interactive quiz application using HTML, CSS, and JavaScript that displays questions, records user responses, tracks the score, shows results, and allows the quiz to be restarted.

## Technologies Used:

- HTML – For creating the structure of the quiz interface
- CSS – For styling the quiz layout and improving user experience
- JavaScript – For quiz logic, interactivity, timer, and result calculation
- Web Browser – To run and test the application

## Explanation

**HTML Structure**

HTML is used to create the layout of the quiz using semantic elements such as <header>, <section>, and <button>.

It displays:

- Quiz title and introduction
- Question text
- Multiple-choice options (radio buttons)
- Next button
- Result section and feedback form

**CSS Styling**

CSS is applied to:

- Improve readability of questions and options
- Align sections and buttons properly
- Use colors, spacing, and borders for better user experience
- Make the quiz interface clean and professional

**JavaScript Functionality**

JavaScript is used to:

- Store questions and answers in an array of objects
- Capture user-selected answers
- Move between questions using button events
- Track and update the score
- Display final results and feedback messages
- Implement a countdown timer for each question
- Allow the quiz to restart

# HTML CODE:

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Smart Quiz App</title>
    <link rel="stylesheet" href="quiz.css">
</head>
<body>

<header>
    <h1>Smart Quiz Challenge</h1>
    <p>Test Your Knowledge in 5 Quick Questions!</p>
</header>

<section class="quiz-container">

    <div id="quiz-box">
        <div id="timer">Time Left: <span id="time">15</span>s
        <div id="progress">Question 1 of 5</div>

        <h2 id="question">Question text</h2>

        <div id="options"></div>

        <button id="next-btn">Next</button>
    </div>

    <div id="result-box" class="hidden">
        <h2>Your Result</h2>
        <p id="score-text"></p>
        <p id="feedback"></p>
        <button id="restart-btn">Restart Quiz</button>
    </div>

</section>

<script src="quiz.js"></script>
```

**CSS CODE:**

```css
* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
    font-family: Arial, sans-serif;
}

body {
    background: linear-gradient(to right, #4e73df, #1cc88a);
    display: flex;
    flex-direction: column;
    align-items: center;
    min-height: 100vh;
    color: #333;
}

header {
    text-align: center;
    padding: 30px;
    color: white;
}

.quiz-container {
    background: white;
    width: 400px;
    padding: 25px;
    border-radius: 10px;
    box-shadow: 0 5px 15px rgba(0,0,0,0.2);
}

#timer {
    font-weight: bold;
    margin-bottom: 10px;
    color: #e74a3b;
}

#progress {
```

```css
#question {
    margin-bottom: 15px;
}

#options label {
    display: block;
    margin-bottom: 10px;
    padding: 8px;
    border: 1px solid #ddd;
    border-radius: 5px;
    cursor: pointer;
    transition: 0.3s;
}

#options label:hover {
    background-color: #f8f9fc;
}

button {
    margin-top: 15px;
    padding: 10px;
    width: 100%;
    border: none;
    background: #4e73df;
    color: white;
    border-radius: 5px;
    cursor: pointer;
    font-size: 16px;
}

button:hover {
    background: #2e59d9;
}

.hidden {
    display: none;
}
```

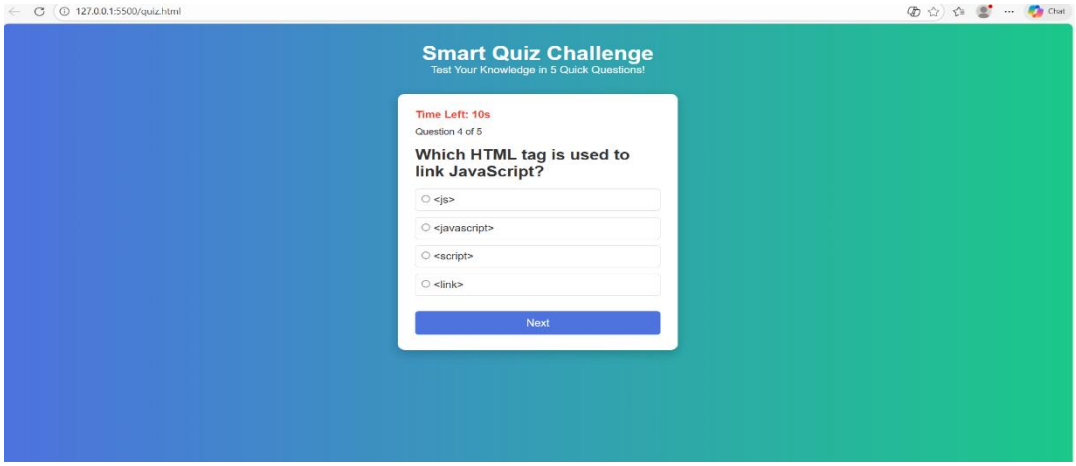
# JAVASCRIPT CODE:

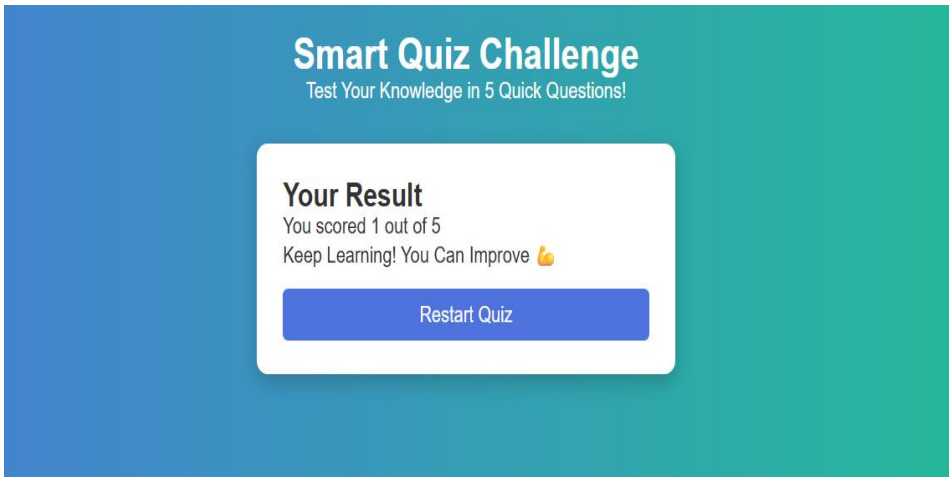
```javascript
const questions = [
    {
        question: "Which language is used for styling web pages?",
        options: ["HTML", "Python", "CSS", "C++"],
        answer: "CSS"
    },
    {
        question: "Which company developed JavaScript?",
        options: ["Microsoft", "Netscape", "Google", "Oracle"],
        answer: "Netscape"
    },
    {
        question: "What does SQL stand for?",
        options: [
            "Structured Query Language",
            "Simple Query Language",
            "System Query Logic",
            "Server Question Language"
        ],
        answer: "Structured Query Language"
    },
    {
        question: "Which HTML tag is used to link JavaScript?",
        options: ["<js>", "<javascript>", "<script>", "<link>"],
        answer: "<script>"
    },
    {
        question: "Which property is used to change text color in CSS?",
        options: ["font-color", "text-color", "color", "background-color"],
        answer: "color"
    }
];

let currentQuestion = 0;
let score = 0;
let timer;
let timeLeft = 15;
```

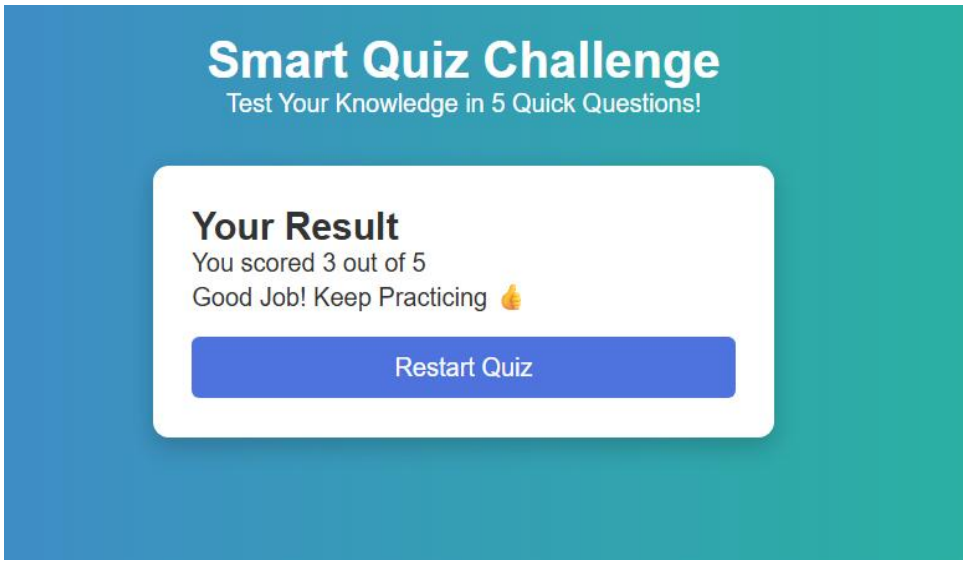
```javascript
const questionEl = document.getElementById("question");
const optionsEl = document.getElementById("options");
const nextBtn = document.getElementById("next-btn");
const timeEl = document.getElementById("time");
const progressEl = document.getElementById("progress");
const resultBox = document.getElementById("result-box");
const quizBox = document.getElementById("quiz-box");
const scoreText = document.getElementById("score-text");
const feedback = document.getElementById("feedback");
const restartBtn = document.getElementById("restart-btn");

function startTimer() {
    timeLeft = 15;
    timeEl.textContent = timeLeft;

    timer = setInterval(() => {
        timeLeft--;
        timeEl.textContent = timeLeft;

        if (timeLeft <= 0) {
            clearInterval(timer);
            nextQuestion();
        }
    }, 1000);
}

function loadQuestion() {
    clearInterval(timer);
    startTimer();

    const q = questions[currentQuestion];
    questionEl.textContent = q.question;
    optionsEl.innerHTML = "";

    progressEl.textContent = `Question ${currentQuestion + 1} of ${questions.length}`;
```

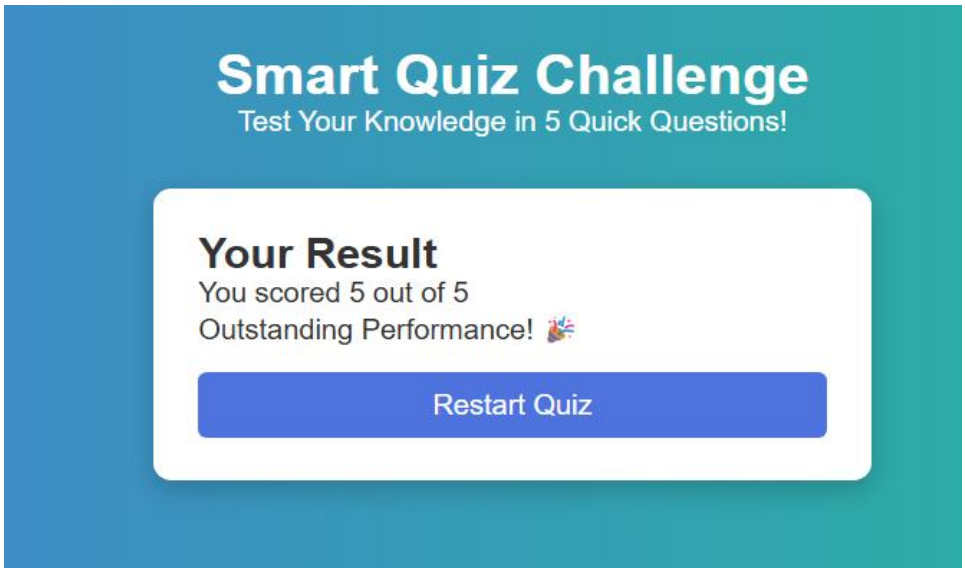
```javascript
function nextQuestion() {
    currentQuestion++;

    if (currentQuestion < questions.length) {
        loadQuestion();
    } else {
        showResult();
    }
}

function showResult() {
    quizBox.classList.add("hidden");
    resultBox.classList.remove("hidden");

    scoreText.textContent = `You scored ${score} out of ${questions.length}`;

    if (score === questions.length) {
        feedback.textContent = "Outstanding Performance! 🎉";
    } else if (score >= 3) {
        feedback.textContent = "Good Job! Keep Practicing 👍";
    } else {
        feedback.textContent = "Keep Learning! You Can Improve 💪";
    }
}

function restartQuiz() {
    currentQuestion = 0;
    score = 0;
    resultBox.classList.add("hidden");
    quizBox.classList.remove("hidden");
    loadQuestion();
}

nextBtn.addEventListener("click", nextQuestion);
restartBtn.addEventListener("click", restartQuiz);

loadQuestion();
```

# OUTPUT:






**RESULT:** An interactive quiz application was successfully created using HTML, CSS, and JavaScript. The application is user-friendly, visually appealing, and meets all the experiment requirements including scoring, result display, feedback, and restart functionality.